



RAPA

*Role-Guided Annotation and Prototype-Aligned Representation Learning
for Historical Literature Sentiment Classification*

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William & Mary Computer Science, with UC San Diego Literature

Structured prompting and low-resource affect

Why this paper matters



Affective labels for text nobody alive learned to read

4

historical corpora —
German, Scandinavian, Latin

+16.7

points of annotation quality
from role-play on Latin

+12.3

macro-F1 on Austrian news
with InfoXLM-large

0

new human annotation —
all labels are LLM-generated

● The claim

Sentiment lexicons do not exist for most historical languages, and LLM labels reflect modern sentiment norms, missing the implicit, ironic and morally nuanced expression typical of historical literature. Fix the first half by making the LLM adopt a historically situated perspective; fix the remaining noise by aligning low-resource representations to stable prototypes learned from high-resource modern data.

● Why this belongs in an affective computing seminar

Every corpus session this term ended at the same place: the labels are the bottleneck. IEMOCAP's kappa of 0.27, CREMA-D's 60.7% agreement, AffectNet's single annotator for 92% of images. This paper takes label noise as the design problem rather than as a caveat.

1

Reading what we have forgotten

Why modern sentiment models fail on historical text



Two problems with the obvious approach



Both concern how LLM labels are produced and used

● Problem 1 — the modern lens

Prompt-based annotation typically uses direct instructions such as 'choose the sentiment of the given text'.

Such strategies implicitly assume that sentiment can be interpreted uniformly across time, culture and context — an assumption that does not hold where sentiment is conveyed implicitly and shaped by rhetorical and moral complexity.

● Problem 2 — assuming the labels are right

Existing work typically assumes the labels produced by LLMs are reliable and uses them directly for downstream training.

But LLM labels are inherently noisy, especially for subtle or stylistically complex expressions. Training on them without denoising limits generalisation.

● And the usual alternative is unavailable

The other branch of prior work uses sentiment lexicons, which are interpretable and easy to apply. But lexicons are often unavailable for historically under-represented texts because the linguistic resources do not exist.

So the paper deliberately avoids lexicons as external knowledge — a constraint, stated as a methodological choice.

Semantic drift



The mechanism behind the modern lens

● What drifts

The sentiment connotations of words change over time. A word that was neutral in its original context may carry positive associations for a modern reader — the paper's example is a term whose modern celebrity associations did not exist when the text was written.

An LLM reading historical text brings its whole training distribution with it.

● And what else changes

Beyond individual words: rhetorical strategies. Irony, indirection, and period-specific norms of sentiment expression.

Classical and Enlightenment literature expresses judgement through virtue vocabulary, metre and allusion — registers a modern sentiment classifier has no handle on.

● The design response

The prompt does not merely ask for a historically informed reading. It explicitly constrains what the model is allowed to know: 'You know nothing of modern persons, events, or ideas.'

That is a negative constraint on the model's knowledge, not a positive instruction about style — and the ablation suggests the constraint is where much of the gain lives.

2

Role-guided annotation

Prompting an LLM to forget the present



The role-play prompt



The Latin example, as printed in the paper

You are a Roman man of letters from the 1st century AD. You live and think entirely within your time. Any word, name, or expression must be interpreted only in the context of Roman knowledge and values.

You know nothing of modern persons, events, or ideas.

Interpret Latin poetry as a Roman would — by meter, style, image, virtue, and feeling, not by the standards of later ages. You classify the overall emotion polarity conveyed by Latin poetic sentences.

Respond ONLY STRICTLY with: positive or negative or neutral or mixed.

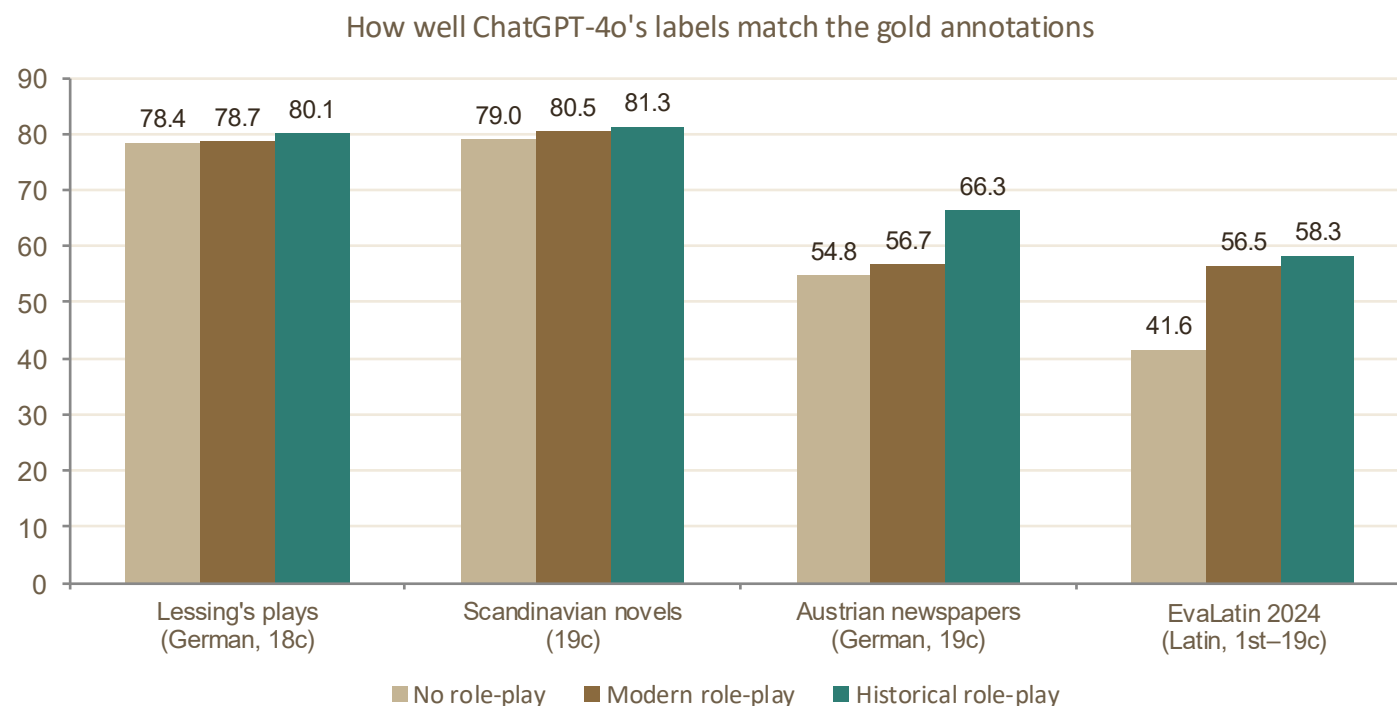
- Three components: a persona, a knowledge constraint, and an interpretive instruction naming period-appropriate criteria — metre, style, image, virtue.
- The knowledge constraint is the distinctive part and appears in every dataset's prompt in some form.
- Prompts are written per dataset. The Lessing prompt casts the model as Lessing himself; the Austrian newspaper prompt as a 19th-century Austrian expert.
- The paper lists this manual prompt design as a limitation: it limits the automation of the annotation process.

Worth noticing what the prompts encode: the Austrian one tells the model not to be influenced by comparative references to political parties or factions. That is historiographical knowledge, written into a prompt by someone who knows the corpus.

Does the role help?



Annotation quality against gold human labels, macro-F1



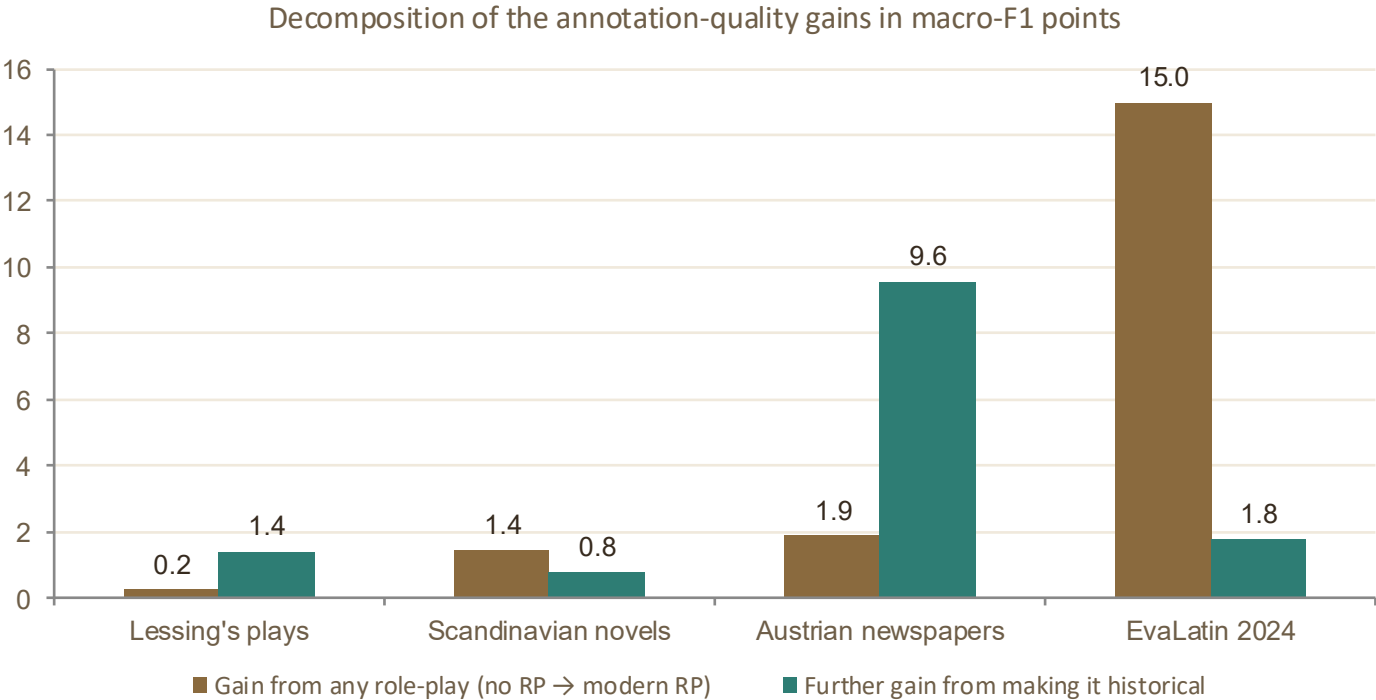
- Historical role-play is best on all four corpora — the paper's headline claim for this component.
- The gains are wildly uneven: +1.6 on Lessing, +2.2 on the novels, +11.4 on Austrian news, +16.7 on Latin.
- The pattern tracks difficulty: the two corpora where plain prompting already works well gain little; the two where it fails gain a lot.
- Note Latin is where a modern GPT-4o has least to go on — the baseline of 41.6 on four classes is close to unusable.

A clean result, well controlled: same model, same texts, same gold labels, only the prompt changes.

How much of the gain is the history?



Decomposing role-play into persona and period



- On Latin, almost all of the gain comes from role-play as such: 15.0 of the 16.7 points arrive with a modern literary critic persona.
- On Austrian newspapers the picture reverses: 9.6 of the 11.4 points require the historical framing specifically.
- So the mechanism differs by corpus. For Latin, adopting any interpretive stance helps; for 19th-century political journalism, the period matters.
- The paper reports both conditions honestly but does not draw this decomposition out.

A plausible reading: the Austrian corpus is full of 19th-century political controversy that a modern reader would score by modern politics. Latin poetry has no such modern analogue to misapply.

3

Prototype-aligned representation learning

Denoising in representation space, not label space



The two-stage framework



Learn anchors from modern data, then pull historical data toward them

Stage 1 — learn prototypes

Train a prototype classifier on high-resource modern data.

A trainable matrix $P_{\text{high}} \in \mathbb{R}^{(C \times d)}$, one row per sentiment class. Classify by softmax over negative squared Euclidean distance from the [CLS] embedding.

Keep the encoder and the prototypes.



Stage 2 — align

Freeze P_{high} as anchors. Initialise low-resource prototypes P_{low} from class means of [CLS] embeddings.

A non-linear mapper M — feedforward with ReLU and LayerNorm — transforms P_{low} into the high-resource space.

Train with cross-entropy plus a symmetric contrastive alignment loss.

● Why this is different from label-level denoising

Prior work identifies noisy labels and corrects or down-weights them before training. This acts on the representation instead: even when a label is wrong, the encoder is pulled toward the prototype its semantic content belongs near.

The paper's framing: existing methods perform label-level denoising in preprocessing but ignore semantic inconsistencies in the representation space.

The two-stage framework

Learn anchors from modern data, then pull historical data toward them

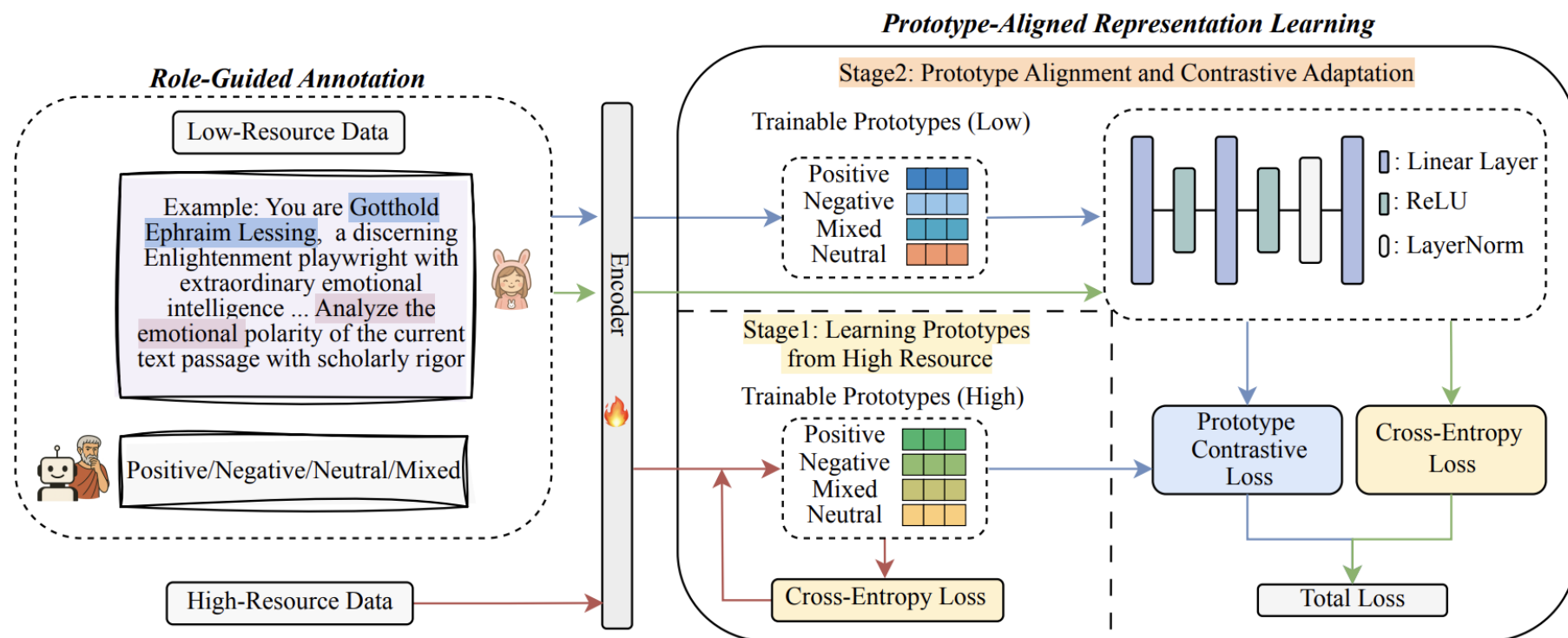


Figure 1: Overall architecture of the proposed RAPA framework. The left side illustrates the role-guided annotation strategy. The right side shows prototype-aligned representation learning. Stage 1 learns high-resource prototypes; Stage 2 performs prototype alignment via contrastive loss and sentiment supervision via cross-entropy loss.

The alignment objective



A symmetric contrastive loss against the identity matrix

● What is being aligned

Not individual examples — class prototypes. The mapped low-resource prototype for 'positive' should match the high-resource prototype for 'positive', and nothing else.

The similarity matrix between the two prototype sets is pushed toward the identity matrix, in both directions.

● Why symmetric

Computing similarities in both directions provides richer supervision signals and promotes one-to-one alignment.

The same instinct as CLIP's symmetric contrastive loss, and as the dual cross-attention in EAA: do not make one side subordinate to the other.

- The total loss is $L = (1 - \alpha) \cdot L_{ce} + \alpha \cdot L_{proto}$, with $\alpha = 0.6$ and temperature $\tau = 0.1$.
- A sweep of α shows 0.6 is best; both 0.3 and 0.9 degrade performance on most datasets — alignment and classification have to be balanced.
- At inference the model uses only the low-resource encoder and mapped prototypes, assigning the nearest prototype in Euclidean space. The high-resource data is not needed.
- That makes deployment lightweight: a multilingual encoder plus four prototype vectors.

The data



Modern corpora as anchors, historical corpora as targets

Role	Dataset	Language	Period	Genre	Size
High-resource	SemEval 2017 Task 4	En / Es / Ar	Contemporary	Twitter	52,850
High-resource	IMDB	English	Contemporary	Film reviews	49,582
High-resource	GoEmotions subset	English	Contemporary	Reddit — supplies 'mixed'	1,021
Low-resource	Lessing's plays	German	18th century	Drama	200 test
Low-resource	Scandinavian novels	Scandinavian	19th century	Fiction	100 test
Low-resource	Austrian newspapers	German	19th century	News	200 test
Low-resource	EvaLatin 2024	Latin	1st–19th century	Poetry	301 test

- The 'mixed' class does not exist in standard sentiment corpora, so GoEmotions sentences containing both positive and negative emotions were relabelled to supply it.
- Lessing and EvaLatin have no training split, so the authors added two further Lessing plays and all 116 Catullus poems, with emotionally salient lines extracted by ChatGPT.
- The test sets are small — 100 to 301 examples — and imbalanced: Austrian news has 96 negative, 17 positive, 79 neutral and 8 mixed.
- Metric is macro-F1 throughout, which is the right choice given that imbalance.

4

Results

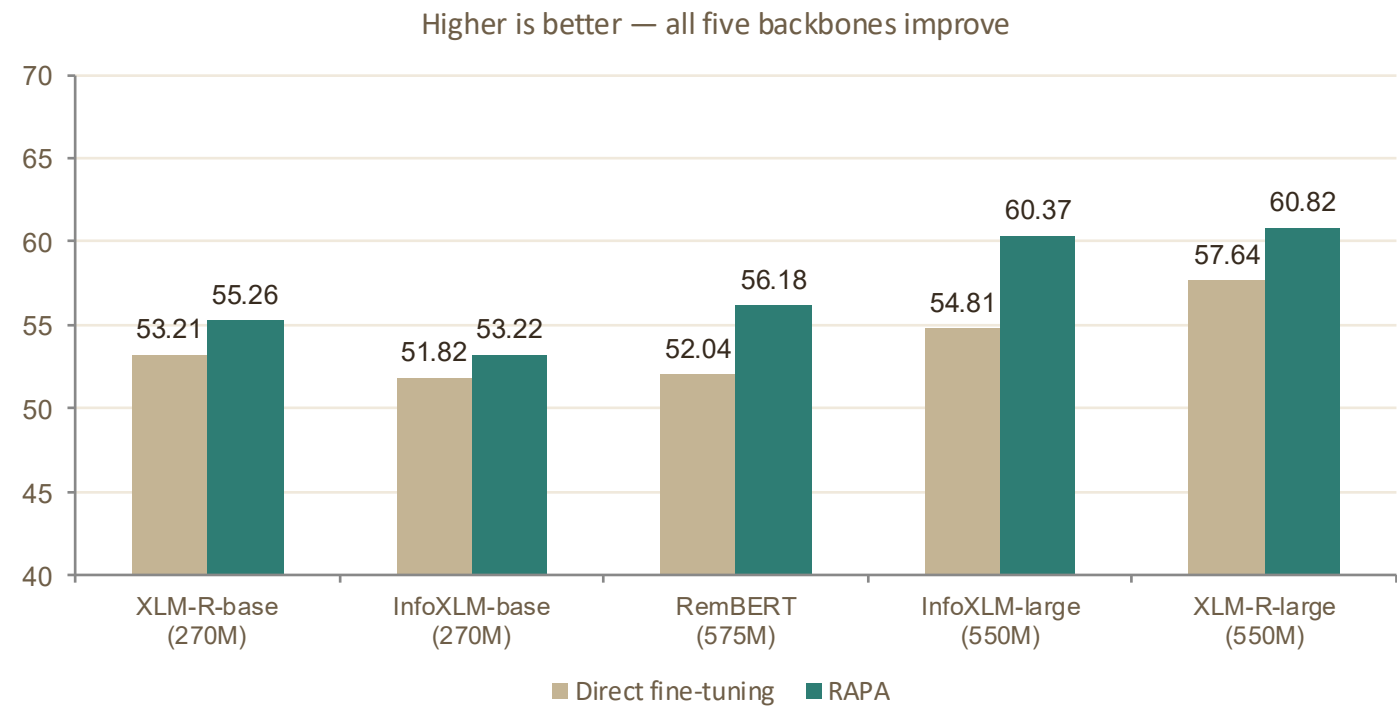
Five backbones, four corpora, and one inversion



Main results



Average macro-F1 across the four historical corpora



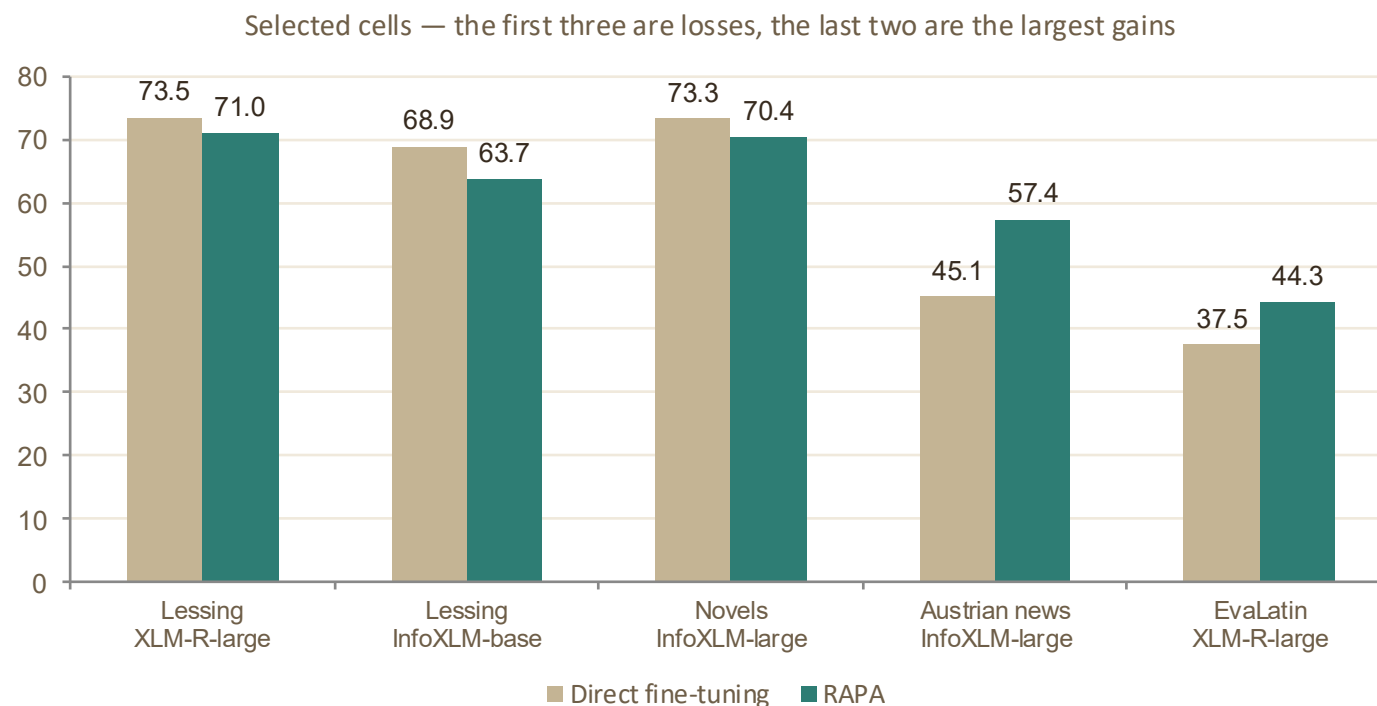
- RAPA improves the average for every backbone, by between 1.4 and 5.6 points.
- The largest gains go to the large models: InfoXLM-large +5.6, RemBERT +4.1, XLNet-large +3.2.
- The single best cell is Austrian newspapers with InfoXLM-large: 45.08 to 57.40, a gain of 12.3.
- On EvaLatin, XLNet-large improves by 6.8 and InfoXLM-large by 7.4 — the denoising working hardest where the labels are noisiest.

Consistency across five encoder architectures is the strongest form of the result — the same argument CCoT-Emo made with three audio LLMs.

Where RAPA does not help



And the paper's explanation



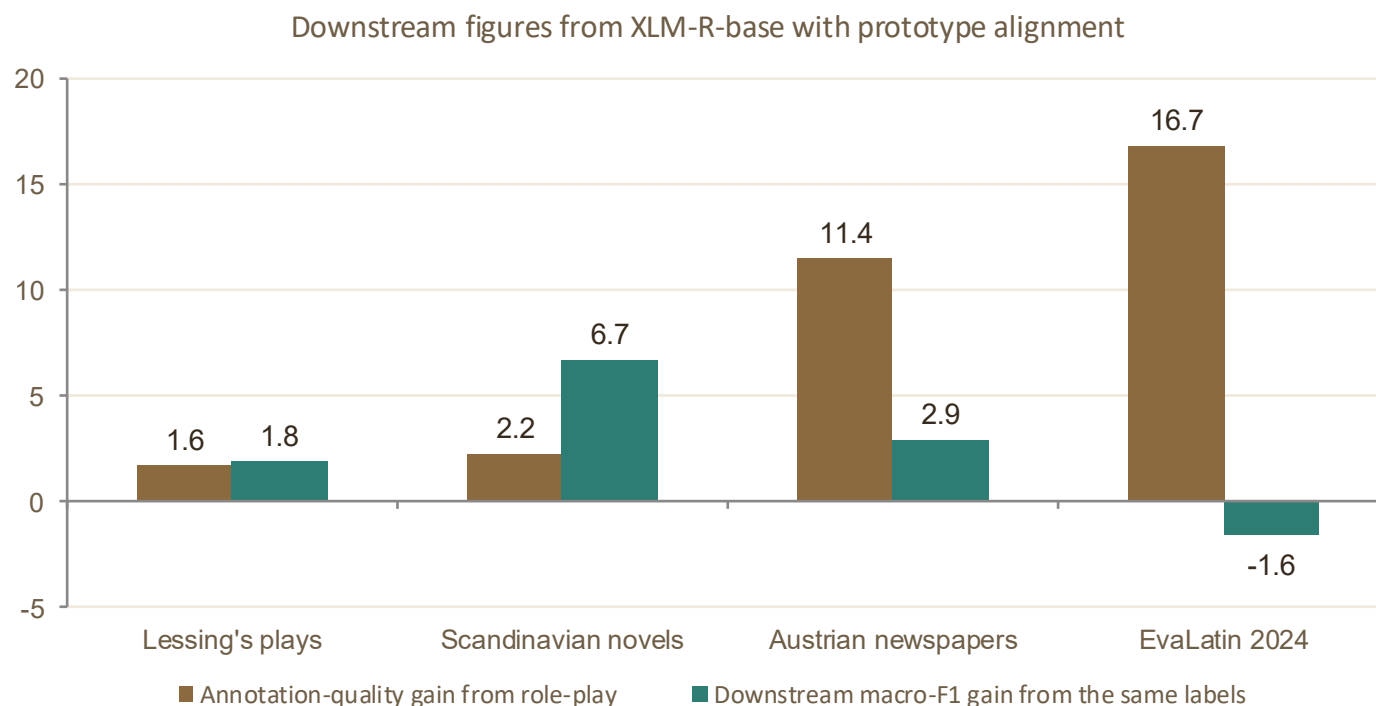
- Lessing is binary — positive and negative only. Scandinavian novels have three classes. Austrian news and EvaLatin have four.
- The paper's hypothesis: a mismatch between the task label space and the prototype space. The framework learns four prototypes, so binary tasks are contrasted against prototypes that do not exist in the task.
- That unnecessary supervision may distort the representation space and hinder class separation.
- The gains line up with the hypothesis: the two four-class corpora gain the most, the binary corpus loses the most.

Credit where due: the paper identifies its own failure pattern, offers a mechanism, and the mechanism predicts the data. That is better than most limitations sections manage.

The inversion



Better labels do not always mean a better model



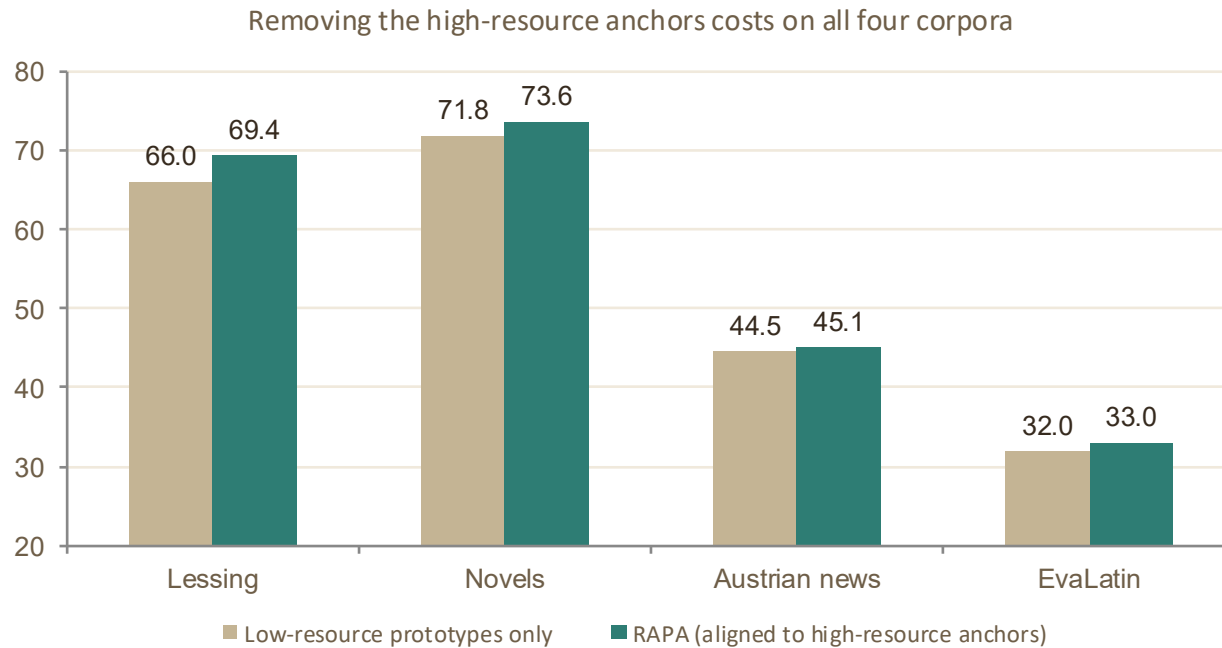
- On EvaLatin, role-play improves annotation quality by 16.7 points and downstream macro-F1 falls by 1.6.
- On Scandinavian novels the reverse: 2.2 points of annotation gain produce 6.7 points downstream.
- On Austrian newspapers, 11.4 points of annotation gain produce 2.9 downstream.
- So there is no reliable relationship between how well the LLM matches gold labels and how well a model trained on those labels performs.

The paper reports both tables and notes they are not directly comparable. Putting them side by side is ours — and it is the most interesting thing in the paper.

Ablations



Two components of the prototype design, on XLM-R-base



- The anchors matter most on Lessing: 3.4 points. The alignment is not decorative.
- On prototype initialisation, trainable beats both mean pooling and first-instance selection — 55.26 against 54.48 and 53.79 on average.
- The representation analysis supports the mechanism: mapping reduces the Euclidean distance between low- and high-resource prototypes for every class, by 3.6 to 7.3 units.
- And the high-resource prototypes are well separated — each is closest to its own class, with 'mixed' showing the tightest intra-class alignment.

• The distance analysis is the right kind of evidence

The claim was that a mapper pulls low-resource prototypes into the high-resource space. Measuring the distances before and after tests that claim directly, rather than inferring it from downstream accuracy.

5

Appraisal and synthesis

The two papers together



Appraisal



What the design buys, and the open threads

● Strengths

A genuine interdisciplinary contribution: the prompts encode real knowledge about how specific corpora should be read, with a literature scholar as co-author.

The annotation ablation is well controlled — same model, same texts, same gold labels, one variable.

Consistent gains across five encoder backbones.

Macro-F1 throughout, which is the right metric for imbalanced affective data and is not what most papers in this series chose.

Self-diagnosis of the failure pattern, with a mechanism that predicts the data.

The representation analysis tests the geometric claim directly rather than inferring it.

● Open threads

Very small test sets — 100 to 301 examples — with no seeds or variance reported. Several reported differences are within plausible run-to-run noise.

The label-space mismatch is diagnosed but not fixed; masking absent classes is one experiment.

Role-play is ablated as a whole, so persona, knowledge constraint and interpretive criteria cannot be separated.

The annotation-quality and downstream gains do not track each other, and the paper does not investigate why.

Prompts are hand-written per corpus, which the authors list as limiting automation.

Four languages and genres; generalisation to other historical corpora is untested.

The two papers together



Structure on the input, structure on the supervision

	CCoT-Emo (ICASSP 2026)	RAPA (Findings of EMNLP 2025)
What is structured	The prompt given to a frozen model	The labels used to train a small model
The structure	An Emotion Graph — attributes and relations as JSON	A role-play persona with a knowledge constraint
What is frozen	Everything — no gradients anywhere	Nothing — a multilingual encoder is fine-tuned
Where human knowledge enters	The choice of seven acoustic features	Hand-written prompts encoding how to read each corpus
Handling noise	Deterministic DSP so the measurement cannot be hallucinated	Representation-level alignment to stable prototypes
Validated across	Three audio LLMs, five benchmarks	Five encoders, four corpora
The dominant channel	Textual attributes — a supervised sentiment classifier	Not decomposed
Both share	Human knowledge in the prompt rather than in the weights	Gains that are large where the baseline is weak

The highlighted row is what makes the pairing worth the session.

Take-aways



Four things to carry out of the session

- 1 Denoise in representation space**

Label-level correction asks whether a label is wrong. Prototype alignment asks where the text belongs regardless — and supervises accordingly.
- 2 Annotator framing is a design variable**

Asking an LLM to read as a first-century Roman is worth 16.7 points of annotation quality on Latin. The prompt is part of the annotation protocol, not a detail of it.
- 3 Better labels are not always a better model**

Annotation gains and downstream gains do not track each other across these four corpora. Consistency may matter more than correctness — the HuBERT lesson, in a different domain.
- 4 Check the label space matches the task**

Aligning a binary task against four prototypes costs points. A mismatch between what you supervise and what you ask for is a real failure mode.

Questions for discussion



- 1 Role-play improves EvaLatin annotation by 16.7 points and downstream macro-F1 falls by 1.6. What explains that, and does it change how you would select annotation strategies?
- 2 On Latin, almost all the annotation gain comes from adopting any persona; on Austrian newspapers, it requires the historical one. What distinguishes the two cases, and what would you predict for other corpora?
- 3 The prompt tells the model 'you know nothing of modern persons, events, or ideas'. Can a language model meaningfully not-know something, or is this a style instruction with an epistemic costume?
- 4 Sentiment prototypes learned from IMDB and Twitter are used as anchors for first-century Latin poetry. What has to be true about sentiment for that transfer to work, and how would you test it?
- 5 Both of today's papers put human expertise into the prompt rather than the weights, against the trend of everything else we read this term. When is that the right bet?